

Lab 3 - Parser

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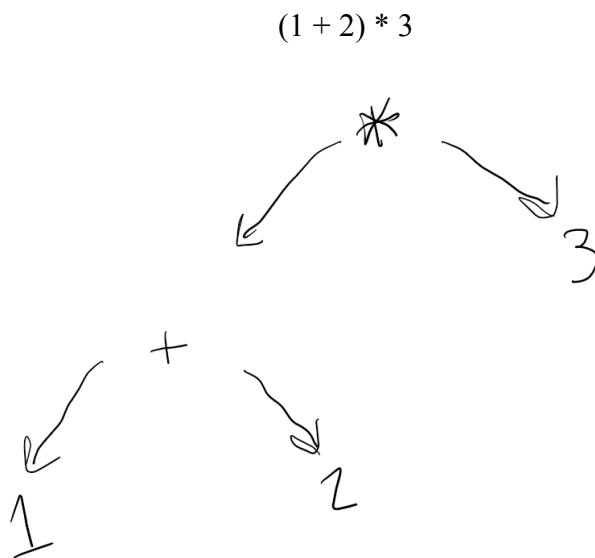
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The goal of this lab was to design and implement a recursive descent parser for the Sloth programming language. The parser is responsible for taking a sequence of tokens, produced by the scanner, and organizing them into a meaningful structure based on the language's grammar. This structure is represented as an abstract syntax tree, which simplifies the raw syntax into a form that reflects the logical relationships between operations and values. In addition to constructing the AST, the lab required integrating the parser with the scanner and the AST printer so that user input could be scanned, parsed, and printed in a parenthesized tree format.

Example Expression:



```
===== SLOTH SHELL =====  
> 1 + 2 + 3  
LINE: 1  
ERROR:  at end: Expect expression.  
PS C:\Users\Blake\OneDrive\Documents\GitHub\CIS343\sloth  
> & C:/Users/Blake/AppData/Local/Microsoft/WindowsApps/p  
ython3.13.exe c:/Users/Blake/OneDrive/Documents/GitHub/C  
IS343/sloth/src/sloth.py  
===== SLOTH SHELL =====  
> 1 * 3 + 2;  
LINE: 1  
ERROR:  at ';': Expect expression.  
PS C:\Users\Blake\OneDrive\Documents\GitHub\CIS343\sloth  
> 
```